MERRYLAND HIGH SCHOOL, ENTEBBE

S.3 MATH HOLIDAY WORK

ITEM ONE

A city's transport department is working on improving traffic and public transport for a busy industrial area where 129 factory workers commute daily. They have conducted a survey to understand the preferred modes of transport used by the workers. The findings have revealed that 32 workers use buses, 50 use motorcycles, and 72 walk to work. Some workers use more than one mode of transport: 15 use both buses and walk, 18 use both buses and motorcycles, and 7 use both motorcycles and walk. 4 workers use only buses, while the number of workers who use none of the three transport modes (they might work remotely or use personal cars) exceeds those who use both motorcycles and walk by 3. The transport planners claim that they will put traffic lights only if the chance of those who only walk exceeds 40%. The transport planners want to analyze this data to optimize road usage and reduce congestion, especially during peak hours.

Task:

a) Determine the number of workers who use all the three modes of transport.

b) Calculate the probability that a randomly selected worker uses at least two of the three modes of

transport.

c) Advise the transport department, based on your calculations, whether to put traffic lights.

ITEM TWO

Your friend owns a rectangular piece of land of which he wants to construct a house with a rectangular cross section leaving a path 2m around the house base. The path occupies an area of 76 m2 and he also wants the dimensions of his house floor to differ by 3m.

He also owns a soft drink depot where he sells Crates of soda and boxes of juice. Tom bought 4 crates of soda and three boxes of juice and paid Shs 110,700. Tom's brother bought 3 crates of soda and 4 boxes of juice and paid Shs 122,400 but he is not contented about the money he paid claiming that he bought almost similar items as Tom but he has paid much money.

Task:

a) (i) Determine the size of the floor base of your friend's house.

(ii) How big is your friend's piece of land?

b) (i) Help Tom's brother to be contented with the payment he made.

(ii) How much would Ann pay if she buys 2 crates of soda and 2 boxes of water from the same depot?

ITEM THREE

You have just got a job at a restaurant in town and the manager has records of the number of customers that asked for a newly introduced dish for the past 40days. The table below shows the number of customers who asked for the new dish for the past 40 consecutive days.

138	145	145	157	150	172	154	140
146	135	128	149	164	147	152	138
168	152	135	125	158	135	168	176
156	150	155	154	126	153	136	163
161	156	144	132	174	160	147	154

The manager has asked you to determine the mean number of customers that ordered for the dish in the past 40days so as to cater for his customer's demands, he also wants to plan well for the ingredients so he asks to determine the highest number of customers that could have asked for the dish in one day for the past 40 days.

Task:

a) How can you use your mathematical skills to determine the mean number of customers that ordered for the dish in the 40days?

b) Using a suitable graph determine the highest number customers that could have ordered for the dish on any of the period stated

ITEM FOUR

Herman and his father Lutaaya have an age difference of 25 years and the product of their ages is 150 years. One day during a mathematics lesson Herman borrows a calculator from his neighbor and on the screen he finds a number 840 which made him wonder which two numbers he could have entered to get the as a product of The figure Later on during the festive season Herman and his elder brother Tom were visited by their uncle and were each given UGX 42,000 and UGX 53,500 Respectively Herman used all his money to buy 4 shirts and 3 vests while Tom used all his money to buy 5 shirts and 4 vests.

Tasks:

(a) How old do you think is Herman and his father Lutaaya?

(b) Help Herman figure out all the possible values that his neighbor could have typed in the in the calculator to get the number on the screen.

c) Help the two brothers explain to their father how much he would spend if he wanted to buy three vests and five shirts for their cousin brother Isaac.

ITEM FIVE

On their Prom party, five students plan to buy their class teacher a gift. The initial plan was that each one buys their own gift to give to the class teacher but later they realized that If they do it as a group would be cheaper So they mobilized themselves and each person contributed as follows;

Student A contributed one hundred thirty -three thousand two hundred fifty shillings.

B contributed 1 1/2 of the amount of A. C contributed 20% more than that of A.

D contributed a fraction which is 1/5 less that of B of the amount of A contributed.

E. She contributed the balance required to make shs 800,000 which was the cost of the gift. On the day of Prom, the money was given to student E to pick the gift from the nearby shop. From school, she used a boda-boda moving at 10km per hour, they moved for 30 minutes due east and then 15 minutes due north to reach the shop.

Tasks:

a). Help to direct another student to locate the shop from school in coordinate form.

b). what amount did each student contributed and who of the five contributed more money.

d). In which other way could they contribute the money more fairly and how much would each contribute that way?